



MRK HOTELS

Review 05 — Laundry Settlement & Room Billing

v1.1.0 · Edition 1.0 · 15 August 2026

MRK Hotels — Review Batch 05 · Laundry Settlement & Room Billing (Problem & Solution)

Module: Laundry (Settlement) — **Status:** Resolved — **5 September 2026**

1. The Problem

Laundry orders had no explicit, controlled **settlement flow**. Any status change could be posted with almost no guardrails, and "post to room" did not verify that the guest was **actually staying in the room** at settlement time.

What this meant in practice:

- A laundry bill could be silently added to a **vacant, dirty or already checked-out room** — creating phantom room charges.
- There was no dedicated UI action saying "*settle this order*" with a conscious choice between **Mark as paid** and **Post to room**.
- Walk-in orders created at the desk (with a room number but no reservation link) could not be posted to the room later without manual intervention.

2. Root Cause

- `applyRoomBilling()` in `LaundryOrderController` posted `room_charges` whenever the order was `billed_to_room`, but:
- it did **not check the reservation's status** (a `checked_out` / vacant stay could still be charged); and
- it required a `reservation_id` that walk-in orders might never have, so their only path was to stay `unpaid` forever.
- The frontend had status buttons but **no settlement modal**, and the payment status was treated like a generic dropdown rather than a deliberate act.

3. The Solution

3.1 Backend — occupied-room-only billing

`applyRoomBilling()` was rewritten to enforce the business rule "**only a room the guest is staying in can be charged**":

1. Only a `payment_status` of `billed_to_room` posts a charge.
2. The order must have `total_charge > 0` (412/422 guard against billing a free order).

3. The reservation is resolved either:
 4. via the order's own `reservation_id` , **or**
 5. for walk-in orders without a link, **by room number** — the latest `confirmed` / `checked_in` reservation on that room (with row-lock for safety).
 6. The reservation must still be an **active stay** (`confirmed` or `checked_in`). Otherwise the request is rejected with **HTTP 422**: *"Only currently occupied rooms can be billed. No active stay found for this room."*
 7. Only then is the charge added to the reservation's `room_charges` .
- Paid and unpaid settlements never touch `room_charges` .

3.2 Frontend — a deliberate settle action

- Unpaid orders now show a **Settle** button in the row actions.
- It opens a **Settlement modal** with a clear choice:
- **Mark as paid** — records the order as settled with cash/card/whatever the hotel uses.
- **Post to room** — adds the charge to the guest's stay. If the order has no reservation link, the modal shows a **room-number field** so a walk-in order can be resolved and posted to the active stay.
- The modal shows the order number, guest and total charge; the backend's **422 rejection message** is surfaced directly in the modal when the target room is not an active stay (e.g. guest already checked out).
- New i18n keys (`settle*`) added to **both** `en.json` and `sw.json` .

4. Files Changed

Area	File	Change
Backend	<code>app/Http/Controllers/Api/V1/Laundry/LaundryOrderController.php</code>	<code>applyRoomBilling()</code> — occupied-room-only, walk-in resolution by room number
Frontend	<code>src/pages/laundry/LaundryListPage.vue</code>	Settle button + settlement modal (paid / post-to-room)
Frontend	<code>src/locales/en.json</code> , <code>sw.json</code>	<code>laundry.settle*</code> keys (both languages)
Tests	<code>tests/Feature/LaundryOrderTest.php</code>	Vacant-room rejection + walk-in settlement tests

5. Verification Results

- `test_billing_to_a_vacant_room_is_rejected` — posting a checked-out reservation to the room returns **422** and the order stays unpaid with **no** room charge.
- `test_walk_in_order_can_be_posted_to_an_occupied_room_at_settlement` — an order created with only `room_number` is resolved to the in-house stay at settlement and the correct `room_charges` are posted.

- `test_settling_an_order_as_paid_needs_no_room` — marking paid needs no reservation at all.
- Backend full suite — **569 passed / 2305 assertions.**
- Frontend `vite build` — **clean**; Vitest **107/107 passed.**

6. Result

Laundry charges can only ever land on a room the guest is actually staying in, walk-in orders can finally be posted to the room when the guest is in-house, and the frontend now treats settlement as an explicit, reviewed action (paid vs. post-to-room) instead of an accidental dropdown flip.